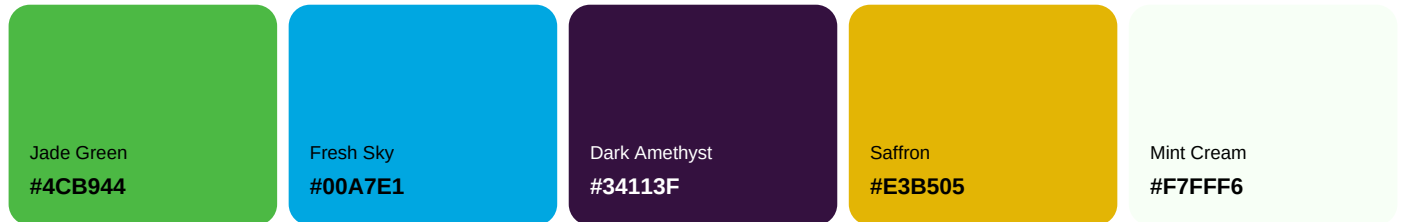


UX in Science Brand Guide

A practical color and accessibility system for web pages, interface components, data graphics, and presentation slides.



Brand posture

UX in Science should feel precise, evidence-driven, calm, and approachable. Use the bright colors to create momentum and wayfinding, but let Dark Amethyst and Mint Cream carry most reading surfaces.

- **Default voice:** clear, concise, research-literate, and human.
- **Visual rhythm:** generous whitespace, strong hierarchy, compact tables, and directly labeled data.
- **Accessibility stance:** color supports meaning; it never carries meaning alone.

Source files

Palette source: `palette.scss`, `palette.txt`, `UX in Science.pdf`, and `Themes-LightDark.png`.

Core Palette

Color	Values	Role	Accessible notes
Jade Green #4CB944	#4CB944 RGB 76, 185, 68 HSL 116, 46%, 50%	Primary accent, positive states, progress	Do not use as small text on light backgrounds. Use as a fill with Dark Amethyst or black text, or use darker text variant #30742B.
Fresh Sky #00A7E1	#00A7E1 RGB 0, 167, 225 HSL 195, 100%, 44%	Secondary accent, links, information	Do not use as small text on light backgrounds. Use as accent fill, underlines, chart markers, or darker link variants #007EA8/#00698C.
Dark Amethyst #34113F	#34113F RGB 52, 17, 63 HSL 286, 58%, 16%	Primary text, headings, strong surfaces	Primary light-theme text and strongest brand surface. Works with Mint Cream, white, Jade, Sky, and Saffron.
Saffron #E3B505	#E3B505 RGB 227, 181, 5 HSL 48, 96%, 45%	Highlight, warnings, callouts	Best for highlights and attention, not body text on light backgrounds. Use #8D7103 or #705A02 for text.
Mint Cream #F7FFF6	#F7FFF6 RGB 247, 255, 246 HSL 113, 100%, 98%	Main light background, quiet panels	Primary light canvas. Pair with Dark Amethyst or black text.

Light and dark theme tokens

Theme	Token	Hex	Usage
Light	Background	#F7FFF6 #F7FFF6	Page or slide canvas.
Light	Text	#34113F #34113F	Body copy, headings, captions.
Light	Primary	#4CB944 #4CB944	Primary accent and positive action.
Light	Info	#00A7E1 #00A7E1	Information accent and chart series.
Light	Highlight	#E3B505 #E3B505	Callouts, notes, and attention.
Dark	Background	#010A00 #010A00	Page or slide canvas.
Dark	Text	#F7FFF6 #F7FFF6	Body copy, headings, captions.
Dark	Primary	#4DBA45 #4DBA45	Primary accent and positive action.
Dark	Info	#1FC7FF #1FC7FF	Information accent and chart series.
Dark	Amethyst accent	#E3BFEE #E3BFEE	Dark-theme tint for secondary emphasis.
Dark	Highlight	#FACE1E #FACE1E	Callouts, notes, and attention.

Accessibility Rules

The guide targets WCAG 2.2 AA contrast: 4.5:1 for normal text, 3:1 for large text and meaningful graphics/UI boundaries, and an additional non-color cue wherever color conveys meaning.

Combination	Ratio	Result	Guidance
Dark Amethyst text on Mint Cream	15.86:1	AAA	Use for body copy and headings in light theme.
Black text on Mint Cream	20.59:1	AAA	Use for utilitarian tables, code, and dense labels.
Jade text on Mint Cream	2.47:1	Fail	Fail for normal text; use #30742B if green text is required.
Fresh Sky text on Mint Cream	2.70:1	Fail	Fail for normal text; use #007EA8 or #00698C for links.
Saffron text on Mint Cream	1.89:1	Fail	Fail for normal text; use #8D7103 or #705A02.
Mint Cream text on Dark Amethyst	15.86:1	AAA	Use for dark cards, banners, and title slides.
Dark Amethyst text on Jade Green	6.42:1	AA	Pass for buttons and badges.
Dark Amethyst text on Fresh Sky	5.87:1	AA	Pass for buttons and badges.
Dark Amethyst text on Saffron	8.38:1	AAA	Pass for buttons, tags, and callouts.
Dark-theme Jade on #010A00	8.06:1	AAA	Pass for dark-mode text, icons, and chart marks.
Dark-theme Sky on #010A00	10.21:1	AAA	Pass for dark-mode links, focus, and chart marks.
Dark-theme Saffron on #010A00	13.32:1	AAA	Pass for dark-mode highlights.

Accessible text variants from the shade scale

Variant	Hex	Contrast on Mint Cream	Use
Jade text	#30742B #30742B	5.63 on Mint Cream	Success labels, status text, data labels.
Sky link	#007EA8 #007EA8	4.53 on Mint Cream	Minimum AA link color on light backgrounds.
Sky strong link	#00698C #00698C	6.06 on Mint Cream	Preferred blue link and chart line.
Saffron text	#8D7103 #8D7103	4.58 on Mint Cream	Minimum AA warning text.
Saffron strong text	#705A02 #705A02	6.53 on Mint Cream	Preferred warning text and chart line.

Web Usage Guidelines

- **Light theme:** use Mint Cream as the page canvas and Dark Amethyst as the default text color. White may be used for cards, tables, and form fields.
- **Buttons and controls:** borderless Jade, Sky, and Saffron components do not meet the 3:1 boundary rule on Mint Cream. Add a 2 px Dark Amethyst border, use a darker shade, or use a Dark Amethyst fill.
- **Links:** do not use base Fresh Sky for small link text on light backgrounds. Use #007EA8 minimum, #00698C preferred, and keep underlines or another persistent cue.
- **Focus states:** use a 2-3 px Dark Amethyst focus ring on light backgrounds. Use dark-theme Sky or Saffron on #010A00.
- **Color meaning:** pair status color with text, icon shape, pattern, or placement. For example, do not communicate required fields or chart categories with color alone.
- **Dark theme:** #010A00 background with Mint Cream text is the default. The dark-theme accent colors are strong enough for text, icons, charts, and focus rings.

Suggested CSS tokens

```
--ux-bg: #F7FFF6;
--ux-surface: #FFFFFF;
--ux-text: #34113F;
--ux-link: #00698C;
--ux-primary: #4CB944;
--ux-primary-text: #34113F;
--ux-info: #00A7E1;
--ux-highlight: #E3B505;
--ux-focus: #34113F;

@media (prefers-color-scheme: dark) {
  --ux-bg: #010A00;
  --ux-surface: #34113F;
  --ux-text: #F7FFF6;
  --ux-link: #1FC7FF;
  --ux-primary: #4DBA45;
  --ux-primary-text: #010A00;
  --ux-info: #1FC7FF;
  --ux-highlight: #FACE1E;
  --ux-focus: #FACE1E;
}
```

Component recipes

Component	Recommended color treatment	Avoid
Primary CTA	Dark Amethyst fill with Mint Cream text, or Jade fill with Dark Amethyst text and Dark Amethyst border.	White text on Jade, Sky, or Saffron.
Status badge	Jade/Sky/Saffron fill, Dark Amethyst text, explicit label, and icon or shape cue.	Color-only status dots.
Form field	White field on Mint Cream with Dark Amethyst label, 1.5-2 px Dark Amethyst or neutral border, clear focus ring.	Pale border that is the only field boundary.
Alert/callout	Saffron header strip or icon with Dark Amethyst text; include alert title and actionable copy.	Saffron paragraph text on light background.

Slides and Data Graphics

- **Default slide:** Mint Cream background, Dark Amethyst title and body text. Use one accent color per slide unless the content is a chart.
- **Dark title slide:** #010A00 background with Mint Cream title and dark-theme accents. Keep dense body copy off dark slides.
- **Text scale:** titles 34-44 pt, body 24-28 pt, annotations at least 18 pt. Keep labels direct and avoid long paragraphs.
- **Scientific images:** avoid text directly over busy images. If overlay is necessary, use an opaque or near-opaque Dark Amethyst/#010A00 panel with Mint Cream text.
- **Projector safety:** test slides in grayscale and on a low-brightness display. Saffron and Mint Cream can wash out if used as thin marks.

Chart palette

Purpose	Light background	Dark background	Guidance
Primary series	Dark Amethyst #34113F	Mint Cream #F7FFF6	Use as the anchor series or baseline.
Green series	Jade -37.5% #30742B	Dark Jade #4DBA45	Use darker Jade for lines/markers on light slides.
Blue series	Sky -37.5% #00698C	Dark Sky #1FC7FF	Preferred for links, methods, secondary datasets.
Yellow series	Saffron -50% #705A02	Dark Saffron #FACE1E	Use with labels or patterns; avoid thin base Saffron lines on light backgrounds.
Accessible category cue	Color plus marker shape, line style, label, or pattern.	Color plus marker shape, line style, label, or pattern.	Do not rely on hue alone for categories or significance.

Slide layout patterns

- Use a Dark Amethyst title bar or left rule when a slide needs more structure.
- Use Saffron sparingly for the single most important conclusion, not every annotation.
- Use Fresh Sky for methods, links, or navigation; use Jade for outcomes, progress, or success.
- For section dividers, use full-bleed Dark Amethyst or #010A00 with Mint Cream text and one accent line.

Production Checklist

- Normal web text reaches at least 4.5:1 contrast; large text and meaningful graphics/UI boundaries reach at least 3:1.
- Interactive controls have visible boundaries and focus indicators that meet 3:1 against adjacent colors.
- Every color-coded status, chart series, or required field has a second cue: text, icon, shape, pattern, or direct label.
- Bright base accents are not used as paragraph text on Mint Cream or white.
- Dark-mode pages use #010A00 background, Mint Cream text, and the dark-theme accent tokens from the theme image.
- Slides remain readable in grayscale and at projector brightness; charts have direct labels and sufficient line weight.

References

WCAG 2.2 SC 1.4.3 Contrast (Minimum): <https://www.w3.org/TR/WCAG22/#contrast-minimum>

WCAG 2.2 SC 1.4.11 Non-text Contrast: <https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast.html>

WCAG 2.2 SC 1.4.1 Use of Color: <https://www.w3.org/WAI/WCAG22/Understanding/use-of-color.html>

Contrast ratios in this guide were computed from the supplied hex values using the WCAG relative luminance formula. The light/dark theme values come from Themes-LightDark.png.